

Measuring the zero resistance of a high-temperature superconductor sample resistance vs. persistent current measurement

We have seen that our four probe method measures the zero resistivity with $\Delta\rho = 6 \times 10^{-8} \Omega\text{m}$ error due to the finite digital resolution. What would be the

τ current decay time constant in our superconducting ring, if its

resistivity was not zero, rather the above $\Delta\rho = 6 \times 10^{-8} \Omega\text{m}$ resistivity error value?

$\tau = L/R$, $R = d\pi\rho/(hw) = 0.125\text{m}\Omega$, $L = \mu_0 A/h = 4\pi \times 10^{-7} \times d^2\pi/(4h) = 1.5 \times 10^{-8} \text{H} \rightarrow \tau = 120\mu\text{s}$

than the four-probe resistance measurement.



Height: $h=15\text{mm}$
Diameter: $d=15\text{mm}$
Wall thickness: $w=1.5\text{mm}$

In case of persistent current measurement, even if the magnetic field measurement is somewhat unstable, we may say, that the current decays by less than 10% within a time of $t=10\text{minutes}=600\text{s}$.

$\rightarrow I(t) = I_0 \exp(-t/\tau) > 0.9 \cdot I_0 \rightarrow \tau > 6000\text{s}$

This calculation shows, that the **persistent current measurement determines the zero resistance with almost 8 orders of magnitude better resolution than the four-probe resistance measurement.**



The magnetic field of a persistent solenoid magnet in a Nuclear Magnetic Resonance (NMR) spectrometer was measured for several years. From this the current decay time-constant of the superconductor turned out to be $\tau > 10^5$ years, with the corresponding resistivity $\rho_{sc} < 10^{-32} \Omega\text{m}!!$

J. File & R. G. Mills, Phys. Rev. Lett 10, 93 (1963)

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Precise temperature control: PID regulation

A simple control task: an object is to be heated to a given T_{setpoint} target (setpoint) temperature by a heating element. The temperature (T) is measured and the heating power (P) is controlled based on the difference between the measured temperature and the setpoint temperature.

Simplest implementation: the operation of a room thermostat:

$$P = \begin{cases} P_0, & \text{if } T_{\text{object}} < T_{\text{setpoint}} \\ 0, & \text{if } T_{\text{object}} > T_{\text{setpoint}} \end{cases}$$

More elaborate, **proportional regulation:**

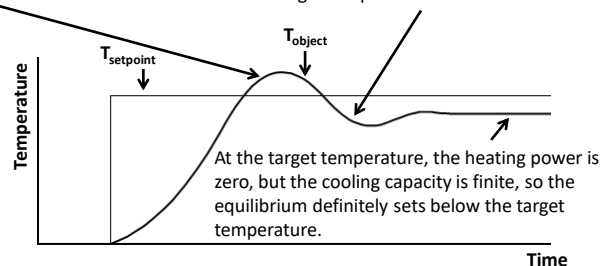
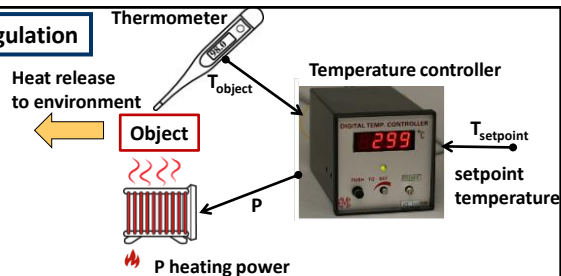
$$P = \begin{cases} C_p \cdot (T_{\text{setpoint}} - T_{\text{object}}), & \text{if } T_{\text{object}} < T_{\text{setpoint}} \\ 0, & \text{if } T_{\text{object}} > T_{\text{setpoint}} \end{cases}$$

Example: the target temperature is raised, how the control reacts:

As the object heats up above the target temperature, the heating is turned off, but the object continues to heat up due to the heat inertia.

If the C_p is too high, the overheating-overcooling oscillation becomes stationary, and no equilibrium is reached.

If the C_p value is too low, we cannot reach the target temperature at all.



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PID control

More precise control: the integral of the error signal is also taken into account
(so-called PI control):

$$\text{output}(t) = C_p(T_{\text{setpoint}} - T_{\text{object}}(t)) + C_I \int_0^t (T_{\text{setpoint}} - T_{\text{object}}(\tau)) d\tau, \quad P = \begin{cases} \text{output, if output} > 0 \\ 0, \text{ if output} < 0 \end{cases}$$

$\nwarrow$
 Proportional constant

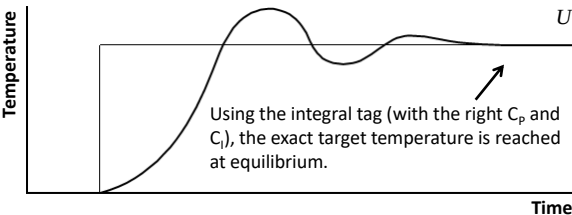
$\nwarrow$
 Integral constant

This is similar to the equation for a series RC circuit:

$$U(t) = R \cdot I(t) + \frac{1}{C} \int_0^t I(\tau) d\tau$$

The voltage of a series RC circuit can change over a characteristic time $\tau_{RC} = RC$. Based on this analogy, the PI control "responds" with a characteristic time of $\tau_{PI} = C_p/C_I$.

The τ_{PI} time constant of the PI controller should be taken to be approximately the same as the time constant of the system due to thermal inertia (see period of oscillation due to thermal inertia)



Note, that in equilibrium ($T_{\text{setpoint}} = T_{\text{object}}$) the integral term is a time-independent (constant) heating power, which actually balances the cooling power of the environment.

The control can be supplemented with a differential term (so-called **PID control**):

$$\text{output}(t) = C_p(T_{\text{setpoint}}(t) - T_{\text{object}}(t)) + C_I \int_0^t (T_{\text{setpoint}}(\tau) - T_{\text{object}}(\tau)) d\tau + C_D \frac{d(T_{\text{setpoint}}(t) - T_{\text{object}}(t))}{dt}$$

$\nwarrow$
 Proportional constant

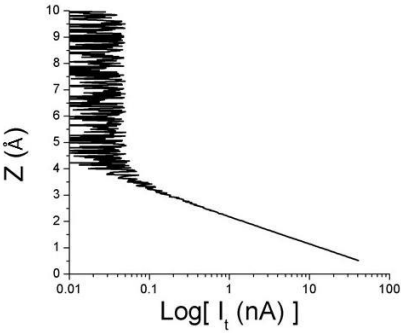
$\nwarrow$
 Integral constant

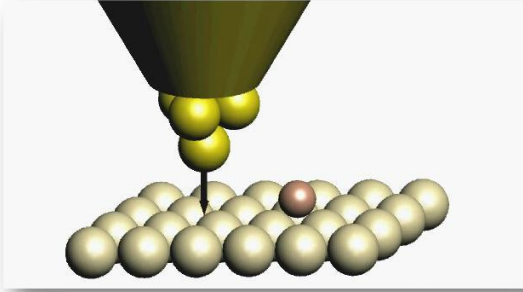
$\nwarrow$
 Differential constant

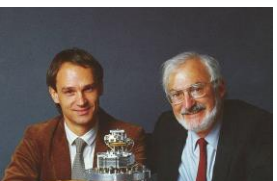
This is useful if the target value also changes over time. In the case of a sudden target value change, the differential term makes the PID control respond faster than if it were only PI control.

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Another example of PI control: scanning tunnelling microscope (STM)

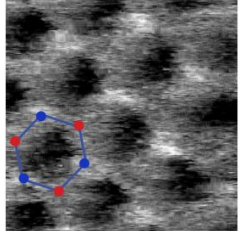






Gerd Binnig, Heinrich Rohrer
Nobel prize 1986.

A quantum mechanical tunnel current flows between a metal tip and a nearby metal surface even when the tip is not touching the surface. The tunnelling current is an exponential function of the tip-sample distance, if the tip is lifted by half an atom-atom distance (~0.1 nm) the tunnelling current decreases by an order of magnitude.

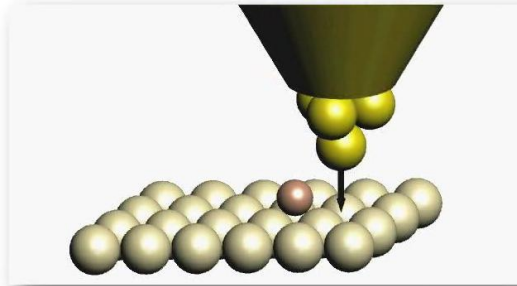
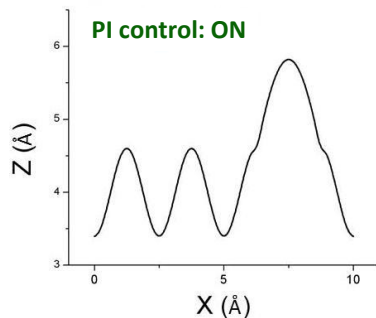


STM image of graphite

„For their design of the scanning tunneling microscope“

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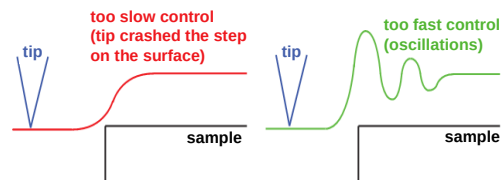
Another example of PI control: scanning tunnelling microscope (STM)



As we sweep the tip over the surface, we raise or lower the tip in the direction perpendicular (z) to the surface so that the tunnel current, i.e. the tip-sample distance, remains the same. This can be achieved using a PI control circuit:

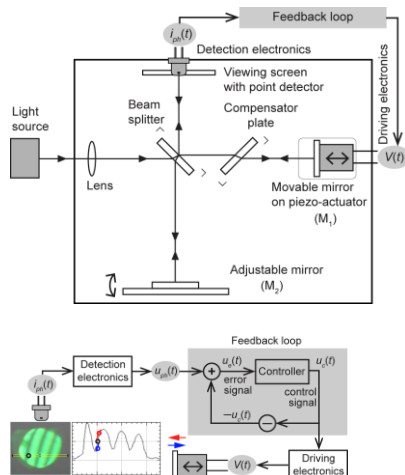
$$\frac{dz}{dt} = C_P(I(t) - I_{\text{setpoint}}) + C_I \int_0^t (I(t) - I_{\text{setpoint}}) dt$$

Here the PI control is not giving the elevation, but the rate of change of elevation (tip velocity).



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Yet another example of PI control: Michelson interferometer



What, if we want to measure larger than wavelength displacements with sub-wavelength resolution?

In normal operation, it is hard to do that, we should count the oscillations also knowing which direction the object moves.

Solution: PI control. The object under study is moving at one mirror. At the other mirror, we move the piezo such that its displacement completely follows the displacement of the studied object. If we succeed, the intensity at the viewing screen will be constant in time.

To do that we actually choose a setpoint intensity halfway between constructive and destructive interference, **we measure the intensity with a photodetector, and we control the piezo actuator such, that the measured intensity is best stabilized at the setpoint intensity.** Like this we can read the displacement of the studied object from the displacement of the feedback-controlled piezo actuator spanning very large displacements with sub-wavelength resolution.

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